

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I P . Sai Charan Tej(192465048)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on

the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE: COMPUTER NETWORKS
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CO2 – ANALYTICAL PROBLEM SOLVING

QUESTION 1

A university network is assigned 192.168.20.0/24. It is divided into Lab A (/26), Lab B (/27), and Lab C (/28). Determine the network address, valid host range, broadcast address, and total usable host addresses for each lab.

1. Problem Understanding

The given network is 192.168.20.0/24. Different fixed-length subnet masks are required for the three labs. The subnets are allocated sequentially from the beginning of the /24 block.

2. Method Selection

Use subnet block size and the formula Usable hosts = $2^{(32 - \text{prefix length})} - 2$. For each subnet, identify the first address as the network address and the last address as the broadcast address.

3. Solution Process

Lab	Prefix	Block Size	Network Address	Valid Host Range	Broadcast	Usable Hosts
Lab A	/26	64	192.168.20.0	192.168.20.1– 192.168.20.62	192.168.20.63	62
Lab B	/27	32	192.168.20.64	192.168.20.65– 192.168.20.94	192.168.20.95	30
Lab C	/28	16	192.168.20.96	192.168.20.97– 192.168.20.110	192.168.20.111	14

4. Accuracy of Calculation

For /26: $2^6 - 2 = 62$ usable hosts. For /27: $2^5 - 2 = 30$ usable hosts. For /28: $2^4 - 2 = 14$ usable hosts.

5. Interpretation of Results

The three labs receive appropriately sized subnets. Lab A supports the largest number of hosts, followed by Lab B and Lab C. The remaining unused range is 192.168.20.112–192.168.20.255.

FINAL ANSWER

Lab A: 192.168.20.0/26, hosts .1–.62, broadcast .63, 62 usable hosts.

Lab B: 192.168.20.64/27, hosts .65–.94, broadcast .95, 30 usable hosts.

Lab C: 192.168.20.96/28, hosts .97–.110, broadcast .111, 14 usable hosts.

QUESTION 2

System 1 has IP 192.168.1.130 with mask 255.255.255.192 and System 2 has IP 192.168.1.190 with the same mask. Determine the subnet range, network address, broadcast address, valid host range, whether the IPs belong to their subnet, and usable hosts.

1. Problem Understanding

The mask 255.255.255.192 is /26. A /26 creates blocks of 64 addresses in the last octet.

2. Method Selection

List the /26 block boundaries: 0, 64, 128, and 192. Locate each IP in the correct block and identify network, host, and broadcast addresses.

3. Solution Process

System	IP Address	Subnet Range	Network	Valid Host Range	Broadcast	Usable Hosts
System 1	192.168.1.130	128–191	192.168.1.128	192.168.1.129– 192.168.1.190	192.168.1.191	62
System 2	192.168.1.190	128–191	192.168.1.128	192.168.1.129– 192.168.1.190	192.168.1.191	62

4. Accuracy of Calculation

Block size = $256 - 192 = 64$. Host bits = 6. Therefore usable hosts = $2^6 - 2 = 62$. Both 130 and 190 fall between 129 and 190, so both are valid host addresses.

5. Interpretation of Results

Both systems belong to the same subnet, 192.168.1.128/26. They can communicate directly at Layer 2 without requiring routing between different IP subnets.

FINAL ANSWER

Both systems are in 192.168.1.128/26. Network = 192.168.1.128; valid hosts = 192.168.1.129–192.168.1.190; broadcast = 192.168.1.191; usable hosts = 62.

QUESTION 3

An organization has 172.16.0.0/16. HR uses /25, Sales /26, IT /27, and Admin /28. Assign suitable subnet addresses and determine network address, host range, broadcast address, usable hosts, and remaining unused IP range.

1. Problem Understanding

The available address block is 172.16.0.0/16. The departments have different subnet sizes. They can be allocated sequentially starting at 172.16.0.0.

2. Method Selection

Use the required prefix length to determine each block size and allocate each subnet at the next valid boundary.

3. Solution Process

Department	Prefix	Network	Valid Host Range	Broadcast	Usable Hosts
HR	/25	172.16.0.0	172.16.0.1– 172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129– 172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193– 172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225– 172.16.0.238	172.16.0.239	14

4. Accuracy of Calculation

The usable host calculations are 126 for /25, 62 for /26, 30 for /27, and 14 for /28.

5. Interpretation of Results

The departments receive different-sized subnets according to their requirements. After the sequential allocation, the remaining unused range is 172.16.0.240–172.16.255.255.

FINAL ANSWER

HR = 172.16.0.0/25; Sales = 172.16.0.128/26; IT = 172.16.0.192/27; Admin = 172.16.0.224/28. Usable hosts are 126, 62, 30, and 14 respectively.

QUESTION 4

A company uses 172.16.10.0/24 and applies /27 subnetting. Determine the total number of subnets, usable hosts per subnet, the subnet containing 172.16.10.78, its network and broadcast addresses, and whether 172.16.10.95 belongs to the same subnet.

1. Problem Understanding

The original prefix is /24 and the new prefix is /27. Therefore, 3 bits are borrowed for subnetting.

2. Method Selection

Number of subnets = $2^{(27-24)}$. Host bits = $32-27$. Block size in the last octet = $256-224 = 32$.

3. Solution Process

- Number of subnets = $2^3 = 8$.
- Usable hosts per subnet = $2^5 - 2 = 30$.
- The /27 ranges are 0–31, 32–63, 64–95, 96–127, 128–159, 160–191, 192–223, and 224–255.

- 172.16.10.78 falls in the 64–95 range.
- Network address = 172.16.10.64.
- Valid host range = 172.16.10.65–172.16.10.94.
- Broadcast address = 172.16.10.95.

4. Accuracy of Calculation

The calculations give 8 subnets and 30 usable hosts per subnet. The address 172.16.10.78 is a valid host in the 64–95 block.

5. Interpretation of Results

172.16.10.95 is within the same subnet boundary, but it is the broadcast address. Therefore it cannot be assigned to a normal host.

FINAL ANSWER

Total subnets = 8; usable hosts/subnet = 30; 172.16.10.78 belongs to 172.16.10.64/27; network = 172.16.10.64; broadcast = 172.16.10.95. The address 172.16.10.95 is the broadcast address of the same subnet and is not a valid host address.

QUESTION 5

A network uses IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Compress it to the shortest form, expand it back to full notation, determine the /64 network prefix, and calculate the total possible host addresses.

1. Problem Understanding

The IPv6 address contains consecutive zero hextets. IPv6 rules allow leading zeros in hextets to be removed and one longest consecutive sequence of zero hextets to be replaced by ::.

2. Method Selection

First remove leading zeros, then compress the longest consecutive zero sequence. For /64, the first four hextets are the network prefix and the remaining 64 bits are available for host/interface addresses.

3. Solution Process

- Given: 2001:0db8:0000:0000:0000:ff00:0042:8329
- After removing leading zeros: 2001:db8:0:0:0:ff00:42:8329
- Compressed form: 2001:db8::ff00:42:8329
- Expanded form: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Network prefix with /64: 2001:0db8:0000:0000::/64
- Host bits = $128 - 64 = 64$
- Total possible addresses = $2^{64} = 18,446,744,073,709,551,616$

4. Accuracy of Calculation

A /64 IPv6 prefix leaves exactly 64 bits after the network portion. Therefore the number of possible addresses is 2^{64} .

5. Interpretation of Results

The compressed IPv6 form is much shorter while representing exactly the same address. The /64 prefix provides a very large address space suitable for a network containing many devices.

FINAL ANSWER

Compressed = 2001:db8::ff00:42:8329. Expanded = 2001:0db8:0000:0000:0000:ff00:0042:8329.

Network prefix = 2001:0db8:0000:0000::/64. Total possible addresses = 2^{64} = 18,446,744,073,709,551,616.

ASSESSMENT FORMAT SUMMARY

Each question is organized according to the Analytical Problem Solving rubric: Problem Understanding (20%), Method Selection (20%), Solution Process (25%), Accuracy of Calculation (20%), and Interpretation of Results (15%).